

Summation Based on Logarithmic Series.

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots$$

$$\log(1-x) = -\left[x + \frac{x^2}{2} + \frac{x^3}{3} + \frac{x^4}{4} + \dots\right]$$

$$\frac{1}{2} \log\left(\frac{1+x}{1-x}\right) = x + \frac{x^3}{3} + \frac{x^5}{5} + \dots$$

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \dots \quad (\text{Gregory's series})$$

$$Q. \quad c \sin \alpha + \frac{c^2}{2} \sin 2\alpha + \frac{c^3}{3} \sin 3\alpha + \dots$$

$$c = c \cos \alpha + \frac{c^2}{2} \cos 2\alpha + \frac{c^3}{3} \cos 3\alpha + \dots$$

$$c + i's = c(\cos \alpha + i \sin \alpha) + \frac{c^2}{2}(\cos 2\alpha + i \sin 2\alpha) + \dots$$

$$= c e^{i\alpha} + \frac{c^2}{2} e^{i2\alpha} + \frac{c^3}{3} e^{i3\alpha} + \dots$$

$$= -\log(1 - c e^{i\alpha})$$

$$= -\log(1 - c(\cos \alpha + i \sin \alpha))$$

$$= -\log(1 - c \cos \alpha - i \sin \alpha \cdot c)$$

$$\text{Let } 1 - c \cos \alpha = r \cos \theta \quad \text{--- (1)}$$

$$c \sin \alpha = r \sin \theta \quad \text{--- (2)}$$

Squaring and adding (1) & (2)

$$r^2 \cos^2 \theta + r^2 \sin^2 \theta = (1 - c \cos \alpha)^2 + (c \sin \alpha)^2$$

$$r^2 = 1 - 2c \cos \alpha + c^2 \cos^2 \alpha + c^2 \sin^2 \alpha$$

$$r^2 = 1 - 2c \cos \alpha + c^2$$

$$r = \sqrt{1 - 2c \cos \alpha + c^2}$$

$$\frac{(2)}{(1)} \Rightarrow \frac{r \sin \theta}{r \cos \theta} = \frac{c \sin \alpha}{1 - c \cos \alpha}$$

$$\tan \theta = \frac{c \sin \alpha}{1 - c \cos \alpha}$$

$$\theta = \tan^{-1} \left(\frac{c \sin \alpha}{1 - c \cos \alpha} \right)$$

$$c + i's = -\log (r \cos \theta - i r \sin \theta)$$

$$= -\log r (\cos \theta - i \sin \theta)$$

$$= -\log r e^{-i\theta}$$

$$= -[\log r + \log e^{-i\theta}]$$

$$= -[\log r - i\theta]$$

$$= -\log r + i\theta$$

$$\Rightarrow \theta = 0$$

$$= \tan^{-1} \left(\frac{c \sin \alpha}{1 - c \cos \alpha} \right)$$

$$Q. c \cos \alpha - \frac{1}{3} c^3 \cos 3\alpha + \frac{1}{5} c^5 \cos 5\alpha - \dots$$

$$S = c \sin \alpha - \frac{1}{3} c^3 \sin 3\alpha + \frac{1}{5} c^5 \sin 5\alpha - \dots$$

$$c + is = c(\cos \alpha + i \sin \alpha) - \frac{1}{3} c^3 (\cos 3\alpha + i \sin 3\alpha) + \dots$$

$$= c e^{i\alpha} - \frac{1}{3} c^3 e^{i3\alpha} + \frac{1}{5} c^5 e^{i5\alpha} - \dots$$

$$= \tan^{-1} c e^{i\alpha}$$

$$\tan(c + is) = c e^{i\alpha}$$

$$\tan(c - is) = c e^{-i\alpha}$$

$$\tan \alpha c = \tan[(c + is) + (c - is)]$$

$$= \frac{\tan(c + is) + \tan(c - is)}{1 - \tan(c + is) \tan(c - is)}$$

$$= \frac{c e^{i\alpha} + c e^{-i\alpha}}{1 - c e^{i\alpha} \cdot c e^{-i\alpha}} = \frac{c(e^{i\alpha} + e^{-i\alpha})}{1 - c^2}$$

$$= \alpha c \frac{(e^{i\alpha} + e^{-i\alpha})}{2} = \alpha c \frac{\cos \alpha}{1 - c^2}$$

$$\tan \alpha c = \frac{2c \cos \alpha}{1 - c^2}$$

$$\alpha c = \tan^{-1} \left(\frac{2c \cos \alpha}{1 - c^2} \right)$$

$$c = \frac{1}{2} \tan^{-1} \left(\frac{2c \cos \alpha}{1 - c^2} \right)$$

$$Q. e^{\alpha} \cos \beta - \frac{1}{3} e^{3\alpha} \cos 3\beta + \frac{1}{5} e^{5\alpha} \cos 5\beta - \dots$$

$$S = e^{\alpha} \sin \beta - \frac{1}{3} e^{3\alpha} \sin 3\beta + \frac{1}{5} e^{5\alpha} \sin 5\beta - \dots$$

$$e + iS = e^{\alpha} (\cos \beta + i \sin \beta) - \frac{1}{3} e^{3\alpha} (\cos 3\beta + i \sin 3\beta) + \dots$$

$$= e^{\alpha} \cdot e^{i\beta} - \frac{1}{3} e^{3\alpha} \cdot e^{i3\beta} + \frac{1}{5} e^{5\alpha} \cdot e^{i5\beta} - \dots$$

$$= \tan^{-1}(e^{\alpha} \cdot e^{i\beta})$$

$$= \tan^{-1}(e^{\alpha + i\beta})$$

$$\tan(C + iS) = e^{\alpha + i\beta}, \quad \tan(C - iS) = e^{\alpha - i\beta}$$

$$\tan 2C = \tan(C + iS + C - iS)$$

$$= \frac{\tan(C + iS) + \tan(C - iS)}{1 - \tan(C + iS) \cdot \tan(C - iS)}$$

$$= \frac{e^{\alpha + i\beta} + e^{\alpha - i\beta}}{1 - e^{\alpha + i\beta} \cdot e^{\alpha - i\beta}}$$

$$\tan 2C = \frac{e^{\alpha} (e^{i\beta} + e^{-i\beta})}{1 - e^{2\alpha}}$$

$$\tan 2C = e^{\alpha} \frac{(2 \cos \beta)}{1 - e^{2\alpha}}$$

$$2C = \tan^{-1} \left(\frac{2e^{\alpha} \cos \beta}{1 - e^{2\alpha}} \right)$$

$$C = \frac{1}{2} \tan^{-1} \left(\frac{2e^{\alpha} \cos \beta}{1 - e^{2\alpha}} \right)$$

HWG. $c \sin \alpha - \frac{c^2}{2} \sin 2\alpha + \frac{c^3}{3} \sin 3\alpha - \dots$ at $\frac{\pi}{2}$ =